

Day 38 — Mon 12 Oct

▶ EXAM DAY — CS Degree Paper-II + MCQ

Countdown: 0 days to CS Degree Paper-II + MCQ

Method: start by recalling yesterday's topic with this page closed. Study the sections below, work every example by hand, then do 20 MCQs.

🕒 Forensic — Paper-I blitz

Source: ComputerForensic/FOR_00_OVERVIEW.md

FOR_00 — Computer Forensic Elective: Overview, Syllabus Checklist & Study Plan

Read this file first, and re-read it every Sunday. It is the map. Everything else is territory.

1. The hard facts

Item	Detail
Exam	Nagaland PSC Combined Technical Services Examination (CTSE) 2026
Elective	Computer Forensic
13 Oct 2026, 09:00–12:00	Technical Descriptive Paper-I — 100 marks, 3 hours
13 Oct 2026, 13:00–15:00	Technical MCQ — 100 questions × 2 = 200 marks , 2 hours
14 Oct 2026, 09:00–12:00	Technical Descriptive Paper-II — 100 marks, 3 hours
Total for this elective	500 marks
Negative marking	MCQ only: −1/3 of the marks per wrong answer (= −0.667)
Days left from 5 Sep 2026	38 days to Paper-I

⚠ Note the brutal shape of 13 October: a 3-hour descriptive paper in the morning and a 2-hour, 100-question MCQ in the afternoon, with roughly a one-hour gap. **Physical stamina is an exam skill.** In the last two weeks do at least three full-length timed simulations that replicate this back-to-back load.

Descriptive paper structure (observed from CTSE 2025 CS-Degree Paper-I; assume identical)

Maximum marks 100			Duration 180 minutes		
Section A	Q1–Q10	answer any 8	8 × 5	=	40
Section B	Q11–Q15	answer any 3	3 × 10	=	30
Section C	Q16–Q19	answer any 2	2 × 15	=	30

Consequence: you get choice in every section. You may legitimately abandon your two weakest topics per section and still score full marks. But choice is only useful if you have *breadth* — a candidate who knows three topics deeply and nothing else will find all three sitting in

Section C and none in Section A.

Time budget inside the descriptive paper

Section	Questions	Marks	Minutes	Per question
A	8 × 5	40	65 min	~8 min
B	3 × 10	30	50 min	~17 min
C	2 × 15	30	55 min	~27 min
Reading / choosing / buffer	—	—	10 min	—

Answer **Section A first** (fastest marks per minute), then C (the long ones, while you are still fresh enough to structure them), then B. Write question numbers clearly; examiners in "answer any N" papers mark the **first N** they find, so cross out anything extra.

2. Where each official unit lives

Legend: **[HERE]** = written in this ComputerForensic folder · **[CORE]** = shared-core file in

`..\_Shared\` written by the shared-core process — do **not** study a second version of it.

⚠ **verify the exact CORE filenames** against `C:\Users\hp\Desktop\prep\Materials\_Shared\` when that process finishes. The names below are the expected ones; if they differ, the *topic* mapping is still correct.

Technical Paper-I

Unit	Official title	Coverage	File
1	Introduction to Forensic Science	[HERE]	<code>FOR_01_ForensicScienceFundamentals.md</code> §1–§3
2	Physical Evidence	[HERE]	<code>FOR_01</code> §4
3	Scene of Crime	[HERE]	<code>FOR_01</code> §5–§6
4	Introduction to Computer Hardware	[HERE]	<code>FOR_02_SystemArtifacts.md</code> §1–§4
5	Windows / Linux / macOS System Artifacts	[HERE]	<code>FOR_02</code> §5–§8
6	Logic, Formula, Unit (Boolean logic, gates, K-maps, flip-flops, number systems)	[CORE]	<code>_Shared\CORE_01_DigitalLogic.md</code> + pointer in <code>FOR_02</code> §9
7	Computer Networks	[CORE]	<code>_Shared\CORE_05_ComputerNetworks.md</code> + forensic slant in <code>FOR_03</code> §3
8	Operating Systems + UNIX + Windows OS	[CORE]	<code>_Shared\CORE_03_OperatingSystems.md</code> , <code>CORE_04_UnixLinux.md</code> + forensic slant in <code>FOR_02</code> §7
9	Computer Organization & Architecture	[CORE]	<code>_Shared\CORE_02_ComputerOrganisation.md</code>

Technical Paper-II

Unit	Official title	Coverage	File
1	Cyber Crime, first responder, search & seizure, imaging & hashing, recovery	[HERE]	FOR_03_CyberCrimeAndIncidentResponse.md §1–§7
2	Web Browsers, Email, VM & Cloud forensics	[HERE]	FOR_03 §8–§10
3	Software Engineering	[CORE]	_Shared\CORE_07_SoftwareEngineering.md
4	Programming in C / OOP / C++ / Web programming	[CORE]	_Shared\CORE_09_C_CPP.md
5	Database & SQL	[CORE]	_Shared\CORE_06_DBMS_SQL.md (SQL injection also in FOR_03 §1)
6	Data & File Structures	[CORE]	_Shared\CORE_08_DataStructures.md
7	Mobile Computing	[HERE]	FOR_04_DigitalEvidenceAndMobile.md §7–§9
8	Digital Evidences	[HERE]	FOR_04 §1–§6

Practice

File	Contents
FOR_05_MCQ_and_QuestionBank.md	120 MCQs with answer key + explanations; 24 × 5-mark, 14 × 10-mark, 10 × 15-mark descriptive questions with model answer skeletons

3. Syllabus tick-box checklist

Print this. Tick a box only when you can reproduce the topic **from memory on blank paper**.

Two passes: [a] = first read done, [b] = recalled from memory.

TP-I Unit 1 — Introduction to Forensic Science

- ☐ Definition of forensic science; forensic vs criminalistics
- ☐ History & development (international milestones; Indian milestones)
- ☐ Scope & development of forensic science **in India**
- ☐ **Locard's Exchange Principle** (state, explain, example)
- ☐ Law of Individuality
- ☐ Principle of Comparison
- ☐ Principle of Analysis
- ☐ Law of Progressive Change
- ☐ Principle of Circumstantial Facts / Facts do not lie
- ☐ Significance & limitations of forensic science
- ☐ **DFSS** — its role and what sits under it
- ☐ **CFSLs** — locations and their specialities
- ☐ **State FSLs, Regional FSLs, Mobile FSLs (MFSU)**
- ☐ **GEQD / CFSL-Kolkata questioned-document work**
- ☐ Divisions inside an FSL and what each does
- ☐ NCRB, BPR&D, NICFS, DNA/Cyber labs — what they are
- ☐ Ethics: code of conduct, expert-witness duty, bias types, ethical failures

TP-I Unit 2 — Physical Evidence

- [] Definition, functions/value of physical evidence
- [] Types (biological, chemical, physical, trace, impression, documentary, electronic)
- [] **Class vs individual characteristics** — definition + 6 examples each
- [] Search methods: strip/line, grid, spiral (in/out), zone/quadrant, wheel/ray — draw all five
- [] **Chain of custody** — meaning, form contents, break consequences

TP-I Unit 3 — Scene of Crime

- [] Meaning; indoor/outdoor; primary/secondary; macroscopic/microscopic
- [] Protection & securing; cordon; the three-layer perimeter; entry/exit log
- [] Documentation: notes, photography (overall→mid→close-up, with/without scale), videography
- [] Sketching: rough vs finished; rectangular coordinate, triangulation, baseline, polar/azimuth
- [] 3-D scanning technique
- [] Processing: discovery → recognition → examination → collection → preservation → packaging → sealing → labelling → forwarding
- [] Correct packaging per evidence type (esp. biological = paper, never plastic)
- [] Safety measures / PPE
- [] Legal: admissibility of e-evidence, **§65B certificate**, IT Act 2000 key sections, **BSA 2023**

TP-I Unit 4 — Computer Hardware

- [] Components, motherboard, chipset, buses; processor; memory hierarchy
- [] HDD geometry: platter, track, sector, cluster, CHS vs LBA
- [] SSD: NAND, wear levelling, TRIM, garbage collection — **the forensic problem**
- [] Boot process: POST → BIOS/UEFI → MBR/GPT → bootloader → kernel → init
- [] File systems: FAT12/16/32, exFAT, NTFS, ext2/3/4, HFS+, APFS — comparison table

TP-I Unit 5 — System Artifacts

- [] Windows registry: hives, file locations, USB history, MRU, Run keys, UserAssist, ShellBags
- [] Event logs + key event IDs
- [] LNK files, prefetch, jump lists
- [] **ADS** — what, why, how to detect
- [] Hidden files & attributes
- [] **File slack vs volume/drive slack** (guaranteed question)
- [] Unallocated space, Recycle Bin (\$I/\$R), thumbcache, pagefile/hiberfil
- [] Linux: layout, inode, permissions, dot-files, /etc/passwd,shadow,group, /var/log, bash_history, cron
- [] macOS: launchd/plists, network config, hidden dirs, unified logs, Keychain, Spotlight, FSEvents

TP-I Units 6–9 — see CORE files

- [] CORE digital logic done
- [] CORE networks done (+ forensic slant read)
- [] CORE OS + UNIX done (+ forensic slant read)
- [] CORE COA done

TP-II Unit 1 — Cyber Crime & Incident Response

- [] Cyber-crime forms & classification; internal vs external attacks
- [] Crimes against person / property / government / society
- [] Social-media crimes; ATM & banking frauds (skimming, cloning, phishing, vishing, UPI)

- [] Data privacy issues; DPDP Act 2023 (awareness level)
- [] Packet sniffing; spoofing (IP/MAC/ARP/DNS/email/caller-ID)
- [] Web security: SQLi, XSS, CSRF, session hijacking, directory traversal, file upload
- [] Malware taxonomy: virus, worm, trojan, ransomware, rootkit, keylogger, botnet, spyware, APT
- [] **First responder** role, toolkit, do's & don'ts, pull-plug vs graceful shutdown
- [] Search & seizure procedure; legal authority
- [] **Order of volatility** (memorise the 8-item list, RFC 3227)
- [] Live vs dead-box forensics
- [] Imaging: bit-stream vs logical copy; write blockers; raw/E01/AFF
- [] Hashing: MD5, SHA-1, SHA-256; verification workflow; collisions
- [] Deleted-file mechanics; **file carving**; **file signatures table**
- [] Steganography & steganalysis
- [] Anti-forensics: wiping, encryption, timestomping, obfuscation, log tampering

TP-II Unit 2 — Browsers, Email, VM/Cloud

- [] Browser artifacts: cookies, bookmarks, cache, history, sessions, plugins, downloads
- [] Storage locations per browser (Chrome/Firefox/Edge/IE)
- [] Private browsing — what survives and where
- [] Email protocols SMTP/POP3/IMAP/MIME; ports; architecture (MUA/MSA/MTA/MDA)
- [] **Header analysis line-by-line** — trace originating IP, detect spoofing (SPF/DKIM/DMARC)
- [] VM forensics: VMDK/VDI/VHD, snapshots, .vmem, memory analysis
- [] Cloud forensics: multi-tenancy, jurisdiction, elasticity, chain of custody, SLA

TP-II Units 3–6 — see CORE files

- [] CORE software engineering done
- [] CORE C/C++/OOP/web done
- [] CORE DBMS/SQL done
- [] CORE data structures done

TP-II Unit 7 — Mobile Computing

- [] Cells, frequency reuse, handoff, cellular framework
- [] Wireless delivery technologies & switching methods
- [] Mobile information access devices
- [] Mobile data internetworking standards; WAP stack
- [] Cellular data protocols 1G→5G, GSM vs CDMA, GPRS/EDGE/UMTS/HSPA/LTE
- [] Mobile computing applications; mobile databases; M-business
- [] Mobile forensics: SIM/handset/card, IMEI/IMSI/ICCID, **CDR analysis**
- [] Acquisition levels: manual → logical → file system → physical → JTAG → chip-off
- [] Android & iOS artifacts; tools

TP-II Unit 8 — Digital Evidence

- [] Storage devices; order of volatility (again)
- [] **ACPO 4 principles**
- [] Digital forensic process model (6 phases) — draw it
- [] Acquisition: physical / logical / sparse; live / static
- [] Data validation; imaging & cloning; hash values
- [] File extensions vs signatures
- [] Registry for evidence (cross-ref)

- [] Data recovery techniques
- [] MBR vs GPT; partitions, volumes, slack
- [] Email tracing; social media analysis; steganalysis
- [] **Forensic report writing + expert testimony**
- [] Tools table (EnCase, FTK, Autopsy, Cellebrite, Wireshark, Volatility, dd, X-Ways)

4. Recommended study order (38 days, ~3.5 h/day)

The ordering principle: **do the forensics-unique material first**, because it is (a) worth the most marks in this elective, (b) entirely new to you, and (c) needs the most repetitions. The CORE material is engineering you already half-know; it revises fast and late.

Phase 1 — Forensics-unique foundation (Days 1–12, 5–16 Sep)

Day	Work
1–2	FOR_01 §1–§3: forensic science definition, history, principles, Indian lab structure , ethics. Make a one-page mind-map of DFSS→CFSL→SFSL→RFSL→MFSU.
3–4	FOR_01 §4–§5: physical evidence, class vs individual, search patterns, chain of custody, crime scene documentation. Hand-draw all 5 search patterns twice.
5	FOR_01 §6: Indian legal framework — §65B, IT Act, BSA 2023. Verify the flagged items.
6–7	FOR_02 §1–§4: hardware, storage geometry, SSD problem, boot process, file system table.
8–10	FOR_02 §5–§6: Windows artifacts — this is the densest, highest-yield block in TP-I. Registry keys and slack space need three passes.
11	FOR_02 §7–§8: Linux + macOS artifacts.
12	Consolidation day. Blank-paper recall of everything in Phase 1. MCQ set 1.

Phase 2 — Cyber crime & digital evidence (Days 13–24, 17–28 Sep)

Day	Work
13–14	FOR_03 §1–§2: cyber-crime taxonomy, frauds, sniffing, spoofing, web attacks, malware.
15–16	FOR_03 §3–§4: first responder, search & seizure, order of volatility, live vs dead. Memorise the volatility list cold.
17–18	FOR_03 §5–§6: imaging, write blockers, hashing, deleted files, carving, signatures, stego, anti-forensics.
19	FOR_03 §7: browser forensics.
20	FOR_03 §8: email header analysis — do the walkthrough twice, then invent your own header and trace it.
21	FOR_03 §9: VM & cloud forensics.
22–23	FOR_04 §1–§6: digital evidence, ACPO, process model, acquisition, MBR/GPT, report writing, tools table.
24	Consolidation. Recall + MCQ set 2 + write two 15-mark answers under time.

Phase 3 — Mobile + CORE sweep (Days 25–32, 29 Sep – 6 Oct)

Day	Work
25–26	FOR_04 §7–§9: mobile computing + mobile forensics + tools.
27	CORE networks revision + FOR_03 §3 forensic slant.
28	CORE OS + UNIX revision + FOR_02 §7 forensic slant.
29	CORE digital logic + COA revision.
30	CORE DBMS/SQL (+ SQL injection link) + software engineering.
31	CORE data structures + C/C++/web programming.
32	Full descriptive mock, Paper-I, 3 hours, timed.

Phase 4 — Drill and simulate (Days 33–38, 7–12 Oct)

Day	Work
33	Full descriptive mock, Paper-II, timed. Self-mark both mocks harshly.
34	Full 100-question MCQ under 2 hours. Analyse every wrong answer.
35	Weak-area repair from mock analysis.
36	Full simulation of 13 Oct: 9–12 Paper-I, break, 13–15 MCQ.
37	Rapid revision: all "Key points" bullet blocks + every table + all diagrams.
38 (12 Oct)	Light. Re-read FOR_00, the volatility list, the signature table, the ACPO principles, the lab structure map. Sleep early.

Daily rhythm (3.5 h)

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0:00–0:20  Active recall of yesterday (blank paper, no notes)
0:20–1:50  New material – read, then immediately convert to bullets in your own words
1:50–2:00  Break
2:00–3:00  Worked examples / diagrams / write one full descriptive answer
3:00–3:30  20 MCQs on today's topic + review

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5. Scoring strategy

MCQ (200 marks, negative 1/3). With -0.667 per wrong and $+2$ per right, a pure random guess among 4 options has expected value $(0.25 \times 2) + (0.75 \times -0.667) = \mathbf{0.0}$ — exactly break-even.

So:

Situation	Action
You know it	Answer.
You can eliminate 1 option (3 left)	$EV = (1/3 \times 2) + (2/3 \times -0.667) = \mathbf{+0.22} \rightarrow$ answer
You can eliminate 2 options (2 left)	$EV = (1/2 \times 2) + (1/2 \times -0.667) = \mathbf{+0.67} \rightarrow$ definitely answer
Zero idea, no elimination	$EV = 0 \rightarrow$ leave it. Not worth the risk or the time.

Practically: **eliminate-then-guess is profitable; blind guessing is not.** Do two passes — pass 1 answer everything you know in ~70 minutes, pass 2 work the eliminations in the

remaining 50. Never leave a question you have eliminated even one option on.

Descriptive. Examiners reward *structure and coverage*, not prose beauty. For every answer:

1. One-line definition in the first sentence (the examiner's first tick).
2. Headed sub-points, numbered or bulleted.
3. A **diagram or table** wherever remotely justified — 15-mark answers without a diagram lose easy marks.
4. An Indian legal/institutional reference where relevant (§65B, IT Act, CFSL, chain of custody form).
5. A one-line conclusion.

Rough length: 5-mark $\approx$ 3/4 page, 10-mark $\approx$ 1.5–2 pages, 15-mark $\approx$ 3 pages + diagram.

6. Things to verify before the exam

Collected list of everything flagged $\triangle$ across these notes — resolve them against a primary source (bare act text on india.gov.in / indiacode.nic.in, DFSS/MHA website) in Phase 3:

1. Exact CORE filenames in `_Shared\`.
2. The current section numbering of the **Bharatiya Sakshya Adhiniyam 2023** for electronic evidence (the BSA replaced the Indian Evidence Act 1872 w.e.f. 1 July 2024; the §65B analogue is generally cited as **BSA §63** with the certificate under **§63(4)** — $\triangle$ verify).
3. The current count and locations of CFSLs (the network has been expanded in recent years, including new National Forensic Sciences University campuses — $\triangle$ verify the current list).
4. Whether NPSC's paper cites the **Indian Evidence Act 1872** or the **BSA 2023** — write both in the exam ("§65B IEA, now §63 BSA 2023") and you cannot be marked wrong.
5. IT Act section numbers beyond the core set (§43, §65, §66, §66C, §66D, §66E, §66F, §67, §69, §70, §72, §79) — do not invent any others.

● Forensic — MCQ bank

Source: ComputerForensic/FOR_05_MCQ_and_QuestionBank.md

FOR_05 — MCQ Bank + Descriptive Question Bank (Computer Forensic, NPSC CTSE 2026)

No CTSE past paper exists for Computer Forensic — this elective is new to the CTSE pattern and no previous-year question paper is available anywhere (official or coaching-circulated). This file is therefore your **main practice source**, built directly from the official syllabus (`OFFICIAL_SYLLABUS.md`, TP-I units 1–9 and TP-II units 1–8) and from `FOR_01–FOR_04`. Treat every question here as a rehearsal for the *shape* the real exam will take, not as a leaked paper. Where a fact is legally/historically uncertain it is marked $\triangle$ **verify** — do not treat those as settled; do not quote them as certainties in the exam without your own check.

0. How to use the negative marking (MCQ paper: $100 \times 2 = 200$, $-1/3$ per wrong answer)

The arithmetic first. Each question is worth **+2** if correct, **-2/3 (-0.667)** if wrong, **0** if left blank.

Strategy	Expected value per question	Verdict
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Leave blank (no guess)	0	Safe floor
Blind guess among 4 options (25% chance correct)	$0.25 \times 2 + 0.75 \times (-0.667) = 0.5 - 0.5 = \approx 0$	Break-even — no better or worse than skipping, pure noise either way
Guess after eliminating 1 wrong option (3 left, 33% chance)	$0.33 \times 2 + 0.67 \times (-0.667) \approx 0.667 - 0.445 = +0.22$	Worth it
Guess after eliminating 2 wrong options (2 left, 50% chance)	$0.5 \times 2 + 0.5 \times (-0.667) = 1 - 0.33 = +0.67$	Strongly worth it
Guess with full confidence in 1 option (no real doubt)	$\approx +2$	Just answer it — this isn't really a "guess"

Rule of thumb for the exam hall:

1. Never blind-guess with zero elimination — it is exactly break-even in theory, but in practice your "gut feel" 4-option guesses tend to run *worse* than 25% on badly-worded distractor questions (this bank is full of paired traps like Locard vs Gross, virus vs worm, MBR vs GPT precisely so you learn to eliminate, not gamble).
2. The instant you can honestly rule out **even one** option, guessing among the rest has positive expected value — do it.
3. Budget your first pass to **answer everything you know cold**, second pass to **eliminate and guess** on partial-knowledge questions, and only leave truly blank the ones where you have *zero* basis to eliminate anything.
4. This syllabus rewards elimination unusually well because so many MCQs are **paired traps** (two named people/two similar acronyms/two similar numbers) — half the option set is usually eliminable on a moment's recall of *which one is which*, even if you can't recall the exact fact being tested.

1. MCQ Bank (127 questions, spanning the whole syllabus)

Numbering is continuous 1–127. Grouped by topic block for study convenience; the real exam will shuffle order and blocks freely, so do not expect this ordering on the paper.

Block A — Forensic Science Fundamentals, Physical Evidence, Crime Scene (Q1–Q20)

Q1. Who is regarded as the "Father of Criminalistics"?

- A) Edmond Locard B) Hans Gross C) Alphonse Bertillon D) Mathieu Orfila

Q2. "Every contact leaves a trace" is a statement of:

- A) Law of Individuality B) Locard's Exchange Principle C) Principle of Comparison D) Law of Progressive Change

Q3. The world's first fingerprint bureau was established in:

- A) Scotland Yard, 1901 B) Calcutta, 1897 C) Shimla, 1904 D) Lyon, 1910

Q4. The first Central Forensic Science Laboratory (CFSL) in India was set up at:

- A) New Delhi B) Hyderabad C) Kolkata (Calcutta) D) Chandigarh

Q5. CFSL New Delhi functions under the administrative control of:

- A) DFSS/MHA B) NCRB C) CBI D) BPR&D

Q6. The Government Examiner of Questioned Documents (GEQD) has its principal office at:

- A) Kolkata B) Hyderabad C) New Delhi D) Shimla

Q7. Which North-East India CFSL is regionally relevant to Nagaland?

- A) CFSL Kolkata B) CFSL Guwahati C) CFSL Bhopal D) CFSL Pune

Q8. NAFIS (fingerprint database) and CCTNS are operated by:

- A) BPR&D B) NCRB C) DFSS D) NFSU

- Q9.** A property shared by an entire manufactured batch/class of objects (e.g., shoe brand and size) is called a:
A) Individual characteristic B) Class characteristic C) Associative evidence D) Transient evidence
- Q10.** Which of these is an example of an individual characteristic?
A) Shoe brand and size B) Blood ABO group C) Striations on a fired bullet from a specific barrel D) Calibre of a firearm
- Q11.** A mismatch in class characteristics between a suspect item and crime-scene evidence results in:
A) Individualisation B) Inconclusive result C) Exclusion of the source D) Automatic conviction
- Q12.** Evidence that is lost quickly if not recorded immediately (odour, temperature, wet stains) is called:
A) Pattern evidence B) Conditional evidence C) Transient evidence D) Associative evidence
- Q13.** A search pattern in which the area is searched twice at 90° to each other is the:
A) Strip search B) Grid search C) Spiral search D) Wheel/ray search
- Q14.** The search pattern best suited to a single body in an open field, searched by one person, is:
A) Zone search B) Grid search C) Spiral search D) Strip search
- Q15.** The main weakness of the wheel/ray search pattern is:
A) It requires many searchers B) Gaps diverge/widen towards the periphery C) It cannot be used outdoors D) It always takes the longest time
- Q16.** The zone/quadrant search pattern is best suited to:
A) Large open fields B) Indoor scenes divided into rooms C) Underwater searches D) Circular blast craters
- Q17.** The documentary record of who held a piece of evidence, when, and what was done to it, from seizure to court, is called the:
A) Panchnama B) Chain of custody C) Forensic linkage triangle D) Order of volatility
- Q18.** "Drylabbing" in forensic ethics means:
A) Testing in a low-humidity lab B) Reporting results of tests never actually performed C) Working without air conditioning D) Refusing to testify
- Q19.** Which bias occurs when an analyst is told irrelevant case facts (e.g., suspect's prior convictions) before comparing evidence?
A) Confirmation bias B) Contextual bias C) Anchoring D) Observer effect
- Q20.** The primary duty of a forensic expert witness in court is to:
A) The party who engaged them B) The investigating officer C) The court D) The victim

Block B — Computer Hardware, OS Basics, File Systems (Q21–Q30)

- Q21.** The component that holds instructions and data currently being processed and loses content on power-off is:
A) ROM B) RAM C) HDD D) SSD
- Q22.** The firmware that performs POST and initiates the boot process on legacy systems is the:
A) BIOS B) UEFI only C) Bootloader D) Kernel
- Q23.** The file system historically native to Windows NT/2000/XP and later, supporting permissions, journaling and large volumes, is:
A) FAT32 B) NTFS C) ext4 D) HFS+
- Q24.** The default file system on modern Linux distributions is typically:
A) NTFS B) exFAT C) ext4 D) APFS
- Q25.** Apple's current file system (successor to HFS+) is:
A) APFS B) exFAT C) ext4 D) ReFS
- Q26.** In the standard boot sequence, which happens first?
A) OS kernel loads B) POST (Power-On Self-Test) C) User login D) Application launch
- Q27.** A motherboard component that houses the CPU is the:
A) DIMM slot B) Processor socket C) PCI slot D) SATA port
- Q28.** RAID is primarily used for:

A) Encrypting data B) Redundancy and/or performance across multiple disks C) Compressing files D) Formatting partitions

Q29. The part of a computer that performs arithmetic and logic operations is the:

A) Control Unit B) ALU C) Cache D) Register file

Q30. Non-volatile memory used to store firmware that is not (normally) rewritten by the user is:

A) RAM B) ROM C) L1 cache D) Virtual memory

Block C — Windows / Linux / macOS Artifacts (Q31–Q45)

Q31. Which Windows feature allows a file to hide additional data streams attached to it, invisible in normal directory listings?

A) Slack space B) Alternate Data Streams (ADS) C) Shadow copy D) Prefetch

Q32. The Windows registry key associated with recording GUI program execution history (values stored ROT-13 encoded) is:

A) UserAssist B) ShellBags C) MountedDevices D) RunMRU

Q33. Windows Registry keys that record USB device connection history (device IDs, first/last connected) are found under:

A) UserAssist B) USBSTOR C) TypedPaths D) WordWheelQuery

Q34. In the NTFS/Windows registry, which of the following carries a LastWrite timestamp?

A) Individual registry values B) Registry keys C) Only the SAM hive D) Only the SYSTEM hive

Q35. Windows ".lnk" shortcut files are forensically significant because they:

A) Are always malicious B) Record evidence of file/program access even if the target is later deleted C) Cannot be analysed D) Only exist on servers

Q36. The unused space between the logical end of a file and the end of the last allocated disk cluster is called:

A) RAM slack / file slack B) Alternate Data Stream C) Unallocated space D) Registry hive

Q37. In Linux, file and directory access rights are primarily controlled through:

A) NTFS ACLs only B) Ownership (user/group/other) and permission bits (read/write/execute) C) Alternate Data Streams D) The Windows Registry

Q38. A hidden file in Linux is conventionally indicated by:

A) A ".hidden" extension B) A leading dot (.) in the filename C) A registry flag D) An ADS marker

Q39. Linux user account and authentication information (hashed passwords) is stored in:

A) /etc/passwd only, in plaintext B) /etc/shadow C) The Windows SAM hive D) /var/log/syslog only

Q40. In macOS forensics, hidden system directories (e.g., containing caches and system state) are analogous in purpose to which Windows artifact category?

A) Prefetch and hidden AppData folders B) The MBR C) The BIOS D) DNS cache only

Q41. macOS system startup and background services are managed primarily through:

A) The Windows Registry B) launchd and property list (.plist) files C) systemd only D) The Bootcamp partition

Q42. Linux system logs recording authentication attempts are typically found in:

A) /var/log/auth.log or /var/log/secure B) C:\Windows\System32\config C) /Applications/Logs D) UserAssist

Q43. Windows Event Logs are forensically valuable primarily because they record:

A) Only network packet captures B) System, security and application events with timestamps (logons, errors, service starts) C) Only browser history D) Only file hashes

Q44. A ".exe" file whose true file signature (magic bytes) begins with "MZ" but has been renamed with a ".jpg" extension is an example of:

A) A legitimate JPEG B) Extension–signature mismatch (masquerading) C) A registry artifact D) An ADS

Q45. Which of the following is NOT typically a Windows artifact category examined in Windows system forensics?

A) Registry B) Event logs C) launchd plist files D) Prefetch files

Block D — Logic, Number Systems, Boolean Algebra, COA basics (Q46–Q55)

Q46. The decimal number 25 in binary is:

A) 11001 B) 11010 C) 10101 D) 11101

Q47. In 2's complement representation, the negative of a binary number is obtained by:

A) Reversing the bit order B) Inverting all bits and adding 1 C) Inverting all bits only (1's complement) D) Adding 1 to the original number

Q48. A logic gate that outputs 1 only when all its inputs are 1 is the:

A) OR gate B) NAND gate C) AND gate D) NOR gate

Q49. A Karnaugh map (K-map) is used to:

A) Encrypt data B) Simplify Boolean expressions C) Design network topologies D) Manage memory paging

Q50. A well-formed formula (WFF) that is true under every possible interpretation is called a:

A) Contradiction B) Tautology C) Contingency D) Satisfiable-only formula

Q51. A circuit whose output depends only on the current inputs (no memory of past state) is a:

A) Sequential circuit B) Combinational circuit C) Flip-flop D) Counter

Q52. A flip-flop is a basic building block of:

A) Combinational circuits only B) Sequential circuits C) ROM only D) The ALU exclusively

Q53. In computer organisation, the memory hierarchy from fastest/smallest to slowest/largest is typically:

A) Main memory → Cache → Registers → Secondary storage B) Registers → Cache → Main memory → Secondary storage C) Secondary storage → Main memory → Cache → Registers D) Cache → Registers → Secondary storage → Main memory

Q54. DMA (Direct Memory Access) is used primarily to:

A) Slow down I/O for accuracy B) Allow I/O devices to transfer data to/from memory without continuous CPU intervention C) Encrypt memory contents D) Replace the ALU

Q55. The hexadecimal representation of decimal 255 is:

A) FF B) F0 C) 1FF D) FE

Block E — Computer Networks (Q56–Q65)

Q56. The OSI model has how many layers?

A) 4 B) 5 C) 7 D) 8

Q57. In the TCP/IP model, IP addressing and routing occur at the:

A) Application layer B) Transport layer C) Internet (network) layer D) Physical layer

Q58. A MAC address is:

A) A logical, changeable software address B) A hardware address burned into the network interface card C) The same as an IP address D) Assigned by DNS

Q59. A device that connects multiple networks and makes forwarding decisions based on IP addresses is a:

A) Hub B) Switch C) Router D) Repeater

Q60. DNS primarily provides the function of:

A) Encrypting web traffic B) Translating domain names to IP addresses C) Assigning MAC addresses D) Managing firewall rules

Q61. A network topology in which all devices connect to a single central cable is called:

A) Star B) Bus C) Ring D) Mesh

Q62. Which transmission medium offers the highest bandwidth and immunity to electromagnetic interference?

A) Twisted pair B) Coaxial cable C) Fibre-optic cable D) Radio wireless

Q63. A firewall's primary function is to:

A) Speed up DNS resolution B) Filter network traffic based on defined security rules C) Store website cookies D) Assign IP addresses dynamically

Q64. Public-key (asymmetric) cryptography differs from secret-key (symmetric) cryptography in that:

A) It uses the same key for encryption and decryption B) It uses a mathematically related key pair — one public, one private C) It cannot be used for encryption, only signing D) It is always faster than symmetric cryptography

Q65. ARP (Address Resolution Protocol) is used to:

A) Resolve domain names to IP addresses B) Resolve IP addresses to MAC addresses on a local network C) Encrypt email D) Route packets across the internet

Block F — Operating Systems Concepts (Q66–Q72)

Q66. Virtual memory allows a system to:

A) Run without any RAM B) Use disk space to extend the effective size of main memory C) Encrypt the file system D) Bypass the CPU scheduler

Q67. The OS technique of dividing memory into fixed-size blocks to reduce external fragmentation is:

A) Segmentation B) Paging C) Swapping D) Caching

Q68. A situation where two or more processes are each waiting for a resource held by the other, and none can proceed, is called:

A) Starvation B) Deadlock C) Thrashing D) Race condition

Q69. Round Robin is a type of:

A) Memory management technique B) CPU scheduling algorithm C) File system D) Network protocol

Q70. Which of these is a malicious program that disguises itself as legitimate software and does not self-replicate?

A) Worm B) Trojan horse C) Virus D) Logic bomb (as commonly distinguished)

Q71. A worm differs from a virus mainly in that a worm:

A) Requires a host file and user execution B) Self-replicates and spreads across networks without needing a host file or user action C) Cannot spread across a network D) Only affects mobile devices

Q72. RAID and disk formatting are primarily discussed under which OS management function?

A) Process management B) Disk management C) Memory paging D) User authentication

Block G — Cyber Crime & First Responder (Q73–Q85)

Q73. Classifying cyber crime by the role of the computer, an attack where the computer itself is the victim (e.g., hacking, DDoS) is called:

A) Computer as tool B) Computer as target C) Computer as incidental D) Computer-associated crime

Q74. Cyber-stalking and identity theft are examples of cyber crime classified against:

A) Property B) The individual/person C) Government D) Society at large

Q75. Compared to external attacks, internal (insider) attacks are generally considered harder to detect because:

A) Insiders never have legitimate access B) Insider activity blends in with normal, already-authorised work C) Insiders always use malware D) Firewalls block insiders automatically

Q76. ARP spoofing operates primarily at which layer/scope?

A) Application layer, across the internet B) Layer 2, within a local broadcast domain C) Physical cabling only D) DNS resolution only

Q77. DNS spoofing/cache poisoning that redirects victims to a fake website is also known as:

A) Phishing B) Pharming C) Smishing D) Vishing

Q78. SQL Injection succeeds primarily because:

A) The web server uses HTTPS B) User input is concatenated unsanitised into an executable SQL query C) The database is encrypted D) DNS records are misconfigured

Q79. Cross-Site Scripting (XSS) differs fundamentally from Cross-Site Request Forgery (CSRF) in that:

A) XSS injects and runs script in the victim's browser; CSRF rides on the victim's already-authenticated session without injecting script B) CSRF injects script; XSS does not C) They are the same attack D) Only CSRF can steal cookies

Q80. The single greatest cause of retail banking/UPI fraud in India is generally attributed to:

A) Advanced technical hacking of bank servers B) Social engineering (phishing/vishing/smishing) tricking victims into sharing OTP/PIN/CVV C) EMV chip cloning D) SIM manufacturing defects

Q81. The first responder's most important immediate duty at a digital crime scene is to:

A) Immediately run antivirus on the machine B) Secure the scene and avoid altering the evidence C) Reboot the

system to check for errors D) Open suspicious files to assess damage

Q82. For a machine found powered OFF at the scene, the correct first-responder action is to:

A) Boot it to check what is on it B) Leave it off and image the disk later using a write blocker C) Immediately connect it to the internet D) Remove the RAM before imaging

Q83. For a machine found powered ON with an unlocked encrypted volume, the recommended priority is to:

A) Pull the plug immediately B) Capture volatile memory / the encryption key first, before any power decision C) Perform a graceful shutdown immediately D) Ignore RAM entirely and proceed straight to disk imaging

Q84. A key risk of a "graceful" (normal) OS shutdown on a compromised machine is that it may:

A) Preserve all volatile evidence B) Trigger shutdown scripts that delete temp files, clear the pagefile, or run anti-forensic wipers C) Always corrupt the file system D) Have no effect on evidence

Q85. To prevent a suspect's phone from being remotely wiped after seizure, the first responder should:

A) Turn on Wi-Fi to back it up B) Place it in a Faraday bag / enable airplane mode and isolate it from the network C) Fully charge it while connected to the internet D) Restart it repeatedly

Block H — Order of Volatility, Imaging & Hashing (Q86–Q93)

Q86. In the order of volatility, which of these should be collected FIRST?

A) Archival backups B) CPU registers and cache C) Hard disk contents D) Printed documents

Q87. In the order of volatility, RAM contents rank as more volatile than:

A) Network connection state / ARP cache B) Registers and cache C) Disk (hard drive) contents D) Nothing — RAM is the most volatile of all

Q88. A hardware write blocker is used during acquisition to:

A) Speed up imaging B) Prevent any write operation from reaching the original evidence disk C) Compress the disk image D) Decrypt the disk

Q89. A "physical" or bit-stream disk image, as opposed to a "logical" acquisition, captures:

A) Only active/visible files B) Every sector including unallocated space and slack — enabling deleted-file recovery C) Only the registry D) Only RAM

Q90. Hashing an acquired forensic image is done primarily to:

A) Compress the image for storage B) Prove and later verify the integrity of the image against the original C) Encrypt the evidence D) Speed up file carving

Q91. Which hash algorithm produces a 256-bit digest and is currently preferred for forensic defensibility over MD5/SHA-1?

A) MD5 B) SHA-1 C) SHA-256 D) CRC32

Q92. MD5 and SHA-1 are considered forensically weaker today mainly because:

A) They are too slow to compute B) Cryptographic collisions have been demonstrated against them C) They cannot be used on Windows D) They produce no fixed-length output

Q93. Imaging a live (running) system's disk without proper care primarily risks:

A) No risk at all — live imaging is always safe B) Altering timestamps and other metadata compared to a dead-box (powered-off) image C) Automatically encrypting the disk D) Deleting the registry

Block I — File Carving, Signatures, Recovery (Q94–Q99)

Q94. File carving is the technique of recovering files by:

A) Reading file system metadata only B) Searching unallocated space for known file signatures/structures, without relying on file system metadata C) Reading the registry D) Reformatting the disk

Q95. The file signature (magic bytes) for a JPEG file typically begins with:

A) 25 50 44 46 B) FF D8 FF C) 50 4B 03 04 D) 4D 5A

Q96. The file signature "MZ" at the start of a file indicates it is (most likely) a:

A) PDF document B) Windows executable (EXE) C) ZIP archive D) PNG image

Q97. A file renamed from ".exe" to ".txt" to disguise it will still be detected as an executable by:

A) Checking only the file extension B) Checking the file's true signature/magic bytes C) Checking only the file's

creation date D) Checking the antivirus name

Q98. On NTFS, when a file is deleted (not securely wiped), typically:

A) The data is instantly and irrecoverably overwritten B) The file's metadata entry is marked free but the underlying data often remains recoverable until overwritten C) The disk is automatically reformatted D) Only the registry is affected

Q99. SSD TRIM commands are forensically significant because they:

A) Improve deleted-file recovery chances B) Can autonomously and irreversibly erase "deleted" data blocks, defeating recovery C) Only affect RAM, not SSDs D) Encrypt all deleted data instead of erasing it

Block J — Browser & Email Forensics (Q100–Q106)

Q100. Small text files stored by websites in a browser to remember session/preference data are called:

A) Bookmarks B) Cookies C) Cache D) Plugins

Q101. Browser cache primarily stores:

A) A list of favourite/bookmarked sites B) Locally saved copies of web page resources for faster reloading C) The user's saved passwords only D) DNS server addresses only

Q102. To trace the true originating server of an email, an investigator reads the "Received:" headers:

A) Top to bottom B) Bottom to top (oldest/originating hop first) C) In random order D) Only the last "Received:" header

Q103. SPF, DKIM and DMARC are email-security mechanisms primarily designed to combat:

A) Slow email delivery B) Email spoofing C) Large attachment sizes D) Spam folders only

Q104. In a spoofed/phishing email, forensic examiners often find a mismatch between the visible "From:" address and the:

A) Subject line B) "Return-Path" / "Reply-To" / originating IP in the headers C) Font used in the body D) Email client version only

Q105. Virtual machine forensics is important because:

A) VMs never contain user data B) A VM can hold an entire independent evidential file system/OS inside a single host, requiring its own acquisition approach C) VMs cannot be imaged D) VMs are identical to physical disks in every respect

Q106. Cloud forensics is distinctively challenging mainly due to:

A) Cloud data is always encrypted and therefore irrelevant B) Data may be distributed across jurisdictions and controlled by a third-party provider rather than the suspect directly C) Cloud providers never retain any logs D) Cloud data cannot be legally obtained under any circumstance

Block K — Digital Evidence, ACPO, Process Model (Q107–Q114)

Q107. Digital evidence is generally described as "latent" because:

A) It never changes B) It is not visible to the naked eye and needs tools/software to be perceived C) It is always encrypted D) It cannot be duplicated

Q108. ACPO Principle 1 states that:

A) An audit trail must be created B) No action taken by law enforcement should change data held on a computer that may later be relied upon in court C) Only senior officers may examine evidence D) All evidence must be destroyed after the case closes

Q109. Under ACPO Principle 2, if an examiner must access original data, they must:

A) Never document their actions B) Be competent and able to explain the relevance and implications of their actions C) Get a second opinion before acting D) Refuse access under all circumstances

Q110. ACPO Principle 4 places overall responsibility for compliance with the principles on:

A) The junior-most officer present B) The person in charge of the investigation C) The defence lawyer D) The manufacturer of the forensic tool

Q111. The six-phase digital forensic process model is typically ordered as:

A) Analysis → Collection → Identification → Preservation → Examination → Presentation B) Identification →

Preservation → Collection → Examination → Analysis → Presentation C) Collection → Identification → Analysis → Presentation → Examination → Preservation D) Presentation → Analysis → Examination → Collection → Preservation → Identification

Q112. NIST SP 800-86 condenses the forensic process to how many core phases?

A) Two B) Three C) Four (Collection, Examination, Analysis, Reporting) D) Six

Q113. Comparing MBR and GPT partitioning, which statement is correct?

A) MBR supports more partitions than GPT B) GPT supports more partitions, larger disks, and keeps a redundant backup table; MBR is limited to ~2 TB and 4 primary partitions C) MBR is used only on macOS D) GPT cannot be used with UEFI

Q114. The legal requirement in Indian evidence law for certifying an electronic record before it is admissible in court is commonly cited as:

A) IT Act §66 B) §65B Indian Evidence Act 1872 / §63 Bharatiya Sakshya Adhiniyam 2023 (△ verify exact current section) C) IT Act §43A D) BNSS §176 only

Block L — Mobile Forensics (Q115–Q122)

Q115. The unique identifier assigned to a mobile handset (hardware) is the:

A) IMSI B) IMEI C) ICCID D) MSISDN

Q116. The identifier stored on the SIM card that identifies the subscriber to the network is the:

A) IMEI B) IMSI C) MAC address D) Serial ATA ID

Q117. A mobile device state where the passcode has not yet been entered since the last full power-on, and full-disk encryption keys are not derived, is called:

A) AFU (After First Unlock) B) BFU (Before First Unlock) C) Airplane mode D) Roaming mode

Q118. To prevent remote wipe and network-based tampering, a seized phone should immediately be placed in a:

A) Anti-static bag only B) Faraday bag C) Standard evidence envelope D) Freezer

Q119. GSM networks use which multiple access technique, and identify subscribers via which component?

A) CDMA; handset-bound identity B) TDMA/FDMA; a removable SIM card C) Only FDMA; no subscriber identity module D) OFDMA; a fixed IP address

Q120. The default on-device database engine used by both Android and iOS to store SMS, call logs, contacts and chat-app data is:

A) MySQL B) SQLite C) Oracle D) MongoDB

Q121. A hard handoff (break-before-make) between cells is characteristic of:

A) CDMA B) GSM C) 5G NR only D) Wi-Fi

Q122. Call Detail Records (CDR) are primarily useful in mobile forensics for:

A) Recovering deleted photos B) Reconstructing call history, duration, and approximate location via cell tower data C) Decrypting the phone's file system D) Bypassing the phone's passcode

Block M — Tools (Q123–Q127)

Q123. A widely used open-source packet capture and network protocol analysis tool is:

A) FTK Imager B) Wireshark C) Volatility D) Autopsy

Q124. A tool commonly used for memory (RAM) forensic analysis to extract running processes and artifacts from a memory dump is:

A) Wireshark B) Volatility C) EnCase Imager only D) NSRL

Q125. FTK Imager and dd/cfdd are primarily used for:

A) Malware reverse engineering B) Disk/memory imaging and acquisition C) Network intrusion detection D) Mobile app development

Q126. The NSRL (National Software Reference Library) hash set is primarily used in forensic examination to:

A) Encrypt evidence B) Filter out (exclude) known, unmodified operating-system and application files by hash, so examiners focus on relevant data C) Crack passwords D) Capture network traffic

Q127. Autopsy is best described as:

A) A network sniffer B) A graphical digital forensics platform (front-end) commonly built on The Sleuth Kit for disk image analysis C) A mobile-only acquisition tool D) A cryptographic hashing algorithm

2. Answer Key with Brief Explanations

Q	Ans	Why
1	B	Hans Gross coined "criminalistics" and wrote the foundational <i>Handbuch für Untersuchungsrichter</i> . Orfila = toxicology; Bertillon = anthropometry/identification; Locard = exchange principle/first crime lab.
2	B	Locard's Exchange Principle — the classic quotable line.
3	B	Calcutta, 1897 (Anthropometric Bureau → Finger Print Bureau) — widely cited as the world's first.
4	C	CFSL Kolkata, 1957 — the oldest CFSL.
5	C	CFSL New Delhi is under the CBI, not DFSS — the classic exam trap.
6	D	GEQD principal office = Shimla; branches at Kolkata and Hyderabad.
7	B	CFSL Guwahati serves the North-East, including Nagaland.
8	B	NCRB runs NAFIS and CCTNS; BPR&D does research/training — don't swap these two.
9	B	Shared by a group/batch → class characteristic.
10	C	Striations are produced by random tool wear unique to one barrel → individual characteristic.
11	C	A class-characteristic mismatch is conclusive of exclusion, even though a match is not conclusive of identity.
12	C	Transient evidence — temporary, must be recorded first (odour, temperature, wet stains).
13	B	Grid search = strip search repeated at 90°; the most thorough pattern.
14	C	Spiral search suits a single searcher around one focal point.
15	B	Wheel/ray search: gaps between spokes widen as you move outward — poor at the periphery.
16	B	Zone/quadrant search maps naturally onto rooms of a building.
17	B	Chain of custody — the documented, unbroken handling record.
18	B	Drylabbing = reporting fictitious/never-performed test results.
19	B	Contextual bias = irrelevant case information colouring the analysis.
20	C	The expert's primary duty is to the court, overriding obligation to the engaging party.
21	B	RAM is volatile working memory.
22	A	BIOS performs POST and starts the legacy boot process (UEFI is the modern equivalent).
23	B	NTFS — Windows NT-family native file system with permissions and journaling.
24	C	ext4 is the common modern Linux default.
25	A	APFS is Apple's current file system, replacing HFS+.
26	B	POST happens first in the boot sequence, before the OS kernel loads.
27	B	The processor socket houses the CPU on the motherboard.

61	B	Bus topology — single central cable/backbone.
62	C	Fibre-optic offers highest bandwidth and immunity to EMI.
63	B	A firewall filters traffic per defined security rules.
64	B	Asymmetric/public-key crypto uses a mathematically related public/private key pair.
65	B	ARP resolves IP addresses to MAC addresses on a local network.
66	B	Virtual memory extends effective RAM using disk space.
67	B	Paging divides memory into fixed-size blocks, reducing external fragmentation.
68	B	Deadlock — circular wait where none of the processes can proceed.
69	B	Round Robin is a CPU scheduling algorithm.
70	B	Trojan horse — disguised, does not self-replicate.
71	B	A worm self-replicates and spreads across networks without needing a host file or user execution.
72	B	RAID/formatting are part of OS disk management.
73	B	Computer-as-target — the machine itself is the victim (hacking, DDoS).
74	B	Cyber-stalking/identity theft target the individual/person.
75	B	Insiders already have legitimate access, so their activity blends with authorised work — harder to detect.
76	B	ARP spoofing operates at Layer 2, within a broadcast domain.
77	B	Pharming = DNS-level redirection to a fake site.
78	B	SQLi succeeds because unsanitised input is concatenated into an executable SQL query.
79	A	XSS injects/executes script in the victim's browser; CSRF abuses an already-authenticated session without injecting script.
80	B	Social engineering (phishing/vishing/smishing) is the dominant cause, not technical hacking.
81	B	Secure the scene and avoid altering evidence — the first responder's top priority.
82	B	If OFF, leave it off; image later with a write blocker (dead-box). Do not boot it.
83	B	Unlocked encrypted volume: capture the key/volatile memory first, or you may permanently lose access.
84	B	Graceful shutdown can trigger scripts that delete temp files/clear the pagefile/run anti-forensic wipers.
85	B	Faraday bag / airplane mode isolates the phone from the network, preventing remote wipe.
86	B	Registers and cache are the most volatile — collect first.
87	C	Order (most→least volatile, relevant excerpt): registers/cache → network/ARP/process tables → RAM → disk. RAM is more volatile than disk.
88	B	A write blocker prevents any write reaching the original evidence disk.
89	B	Physical/bit-stream imaging captures every sector, including unallocated space and slack.
90	B	Hashing proves and later verifies integrity of the image against the original.
91	C	SHA-256 produces a 256-bit digest and is currently preferred over MD5/SHA-1.

92	B	MD5/SHA-1 are weakened by demonstrated cryptographic collisions.
93	B	Live imaging without care alters timestamps/metadata compared to a dead-box image.
94	B	File carving searches unallocated space for known signatures/structures, independent of file system metadata.
95	B	JPEG signature begins FF D8 FF.
96	B	"MZ" (4D 5A) signature = Windows executable.
97	B	Signature/magic-byte analysis, not the extension, reveals the true file type.
98	B	Deleting a file (without secure wipe) frees the metadata entry but the data often remains recoverable until overwritten.
99	B	SSD TRIM can autonomously and irreversibly erase "deleted" blocks, defeating recovery.
100	B	Cookies — small text files storing session/preference data.
101	B	Browser cache stores locally saved copies of page resources for faster reloading.
102	B	Read "Received:" headers bottom to top to trace back to the originating hop.
103	B	SPF/DKIM/DMARC are anti-email-spoofing mechanisms.
104	B	Mismatch between visible "From:" and Return-Path/Reply-To/originating IP reveals spoofing.
105	B	A VM can be an entire independent evidential OS/file system inside one host file, needing its own acquisition approach.
106	B	Cloud data may span jurisdictions and sit under third-party provider control, not the suspect's direct custody.
107	B	"Latent" = not visible to the naked eye; needs tools to perceive.
108	B	ACPO Principle 1 — no action should change data that may be relied upon in court.
109	B	ACPO Principle 2 — competent access with documented relevance/implications, when unavoidable.
110	B	ACPO Principle 4 — overall responsibility rests with the person in charge of the investigation.
111	B	Identification → Preservation → Collection → Examination → Analysis → Presentation.
112	C	NIST SP 800-86 condenses to four phases: Collection, Examination, Analysis, Reporting.
113	B	GPT: more partitions, larger disks, redundant backup table; MBR: ~2 TB limit, max 4 primary partitions.
114	B	§65B Indian Evidence Act 1872 / §63 Bharatiya Sakshya Adhiniyam 2023 — the electronic-record certificate requirement (△ verify exact current section before quoting in the exam).
115	B	IMEI identifies the handset (hardware).
116	B	IMSI identifies the subscriber, stored on the SIM.
117	B	BFU (Before First Unlock) — passcode not yet entered since power-on; encryption keys not yet derived.
118	B	Faraday bag blocks radio signals, preventing remote wipe/network tampering.
119	B	GSM uses TDMA/FDMA and identifies subscribers via a removable SIM card.
120	B	SQLite is the default on-device database engine on both Android and iOS.

121	B	Hard handoff (break-before-make) is characteristic of GSM; CDMA uses soft handoff.
122	B	CDRs reconstruct call history/duration and approximate location via cell tower data.
123	B	Wireshark — packet capture and protocol analysis.
124	B	Volatility — memory (RAM) forensic analysis framework.
125	B	FTK Imager and dd/dcfldd are disk/memory imaging and acquisition tools.
126	B	NSRL hash sets exclude known-good OS/application files by hash, letting examiners focus on relevant data.
127	B	Autopsy is a graphical forensics platform commonly built atop The Sleuth Kit.

3. Descriptive Question Bank

Structured to match the real descriptive paper shape observed for CTSE ([OFFICIAL_SYLLABUS.md §1](#)): **Section A — any 8 of 10 × 5 marks; Section B — any 3 of 5 × 10 marks; Section C — any 2 of 4 × 15 marks**, per paper (Paper-I and Paper-II are each set out this way — you get *choice*, so learn breadth over depth: skip your two weakest topics per section on the real paper). Each question below carries a **model-answer skeleton** — the bullet points an examiner is checking for, not full prose. Write the prose yourself in practice; do not memorise sentences verbatim.

3.1 Five-mark questions (22 — Section A style)

D1. State and explain Locard's Exchange Principle with two examples.

- Definition: mutual exchange of material whenever two objects contact.
- One physical example (burglar/window: fibres left, glass taken).
- One digital analogue (opening a file leaves MFT/prefetch/LNK/registry traces).
- Note: absence of detectable trace ≠ absence of contact (detection depends on time/medium/skill).

D2. Distinguish class characteristics from individual characteristics with examples.

- Definitions (group-associable vs single-source-associable).
- Table/list: 3 examples of each (shoe brand vs wear pattern; blood group vs DNA STR; calibre vs striations).
- Key line: class mismatch = exclusion; class match ≠ identity.

D3. What is forensic science? Discuss its scope in India.

- Definition (application of science to legal questions).
- 5–6 domains with one example question each (physics, chemistry, toxicology, biology/serology, ballistics, QD, fingerprints, digital).
- One line on growth drivers (cybercrime rise, BNSS mandate) and one constraint (backlog).

D4. Write a note on the Directorate of Forensic Science Services (DFSS).

- Parent: MHA; created 2002; HQ New Delhi.
- Controls CFSs and GEQD.
- Functions: standards/SOPs, referral lab, training, R&D, advice to MHA.

D5. What is a Mobile Forensic Science Unit (MFSU)? State its functions.

- Definition: vehicle-based lab reaching the crime scene.
- Kit: crime-scene/photography equipment, presumptive tests, collection/packaging materials.
- Purpose: immediate scene examination, prevent contamination/loss, link to BNSS mandatory-forensics requirement.

D6. What is confirmation bias? How can a laboratory minimise it?

- Definition: seeing what one expects to see.
- One example (told of a confession, then "finds" a match).
- Countermeasures: Linear Sequential Unmasking, context management, blind verification, accreditation.

D7. Distinguish between the Central and State Forensic Science Laboratories.

- Table: control (MHA/DFSS vs State Home Dept/Police), scope (central agencies + referral vs State casework), examples (CFSL Kolkata/Guwahati vs State FSL).

D8. Distinguish MBR and GPT.

- Table: firmware era (BIOS/UEFI), location (LBA0 vs LBA1+backup), max partitions (4 vs 128), max size (~2TB vs huge), redundancy (none vs backup+CRC32).

D9. State the ACPO principles of digital evidence.

- List all four, one line each (no change; competent+documented access; audit trail/reproducibility; overall responsibility with person in charge).

D10. List the characteristics of digital evidence.

- Latent, fragile/volatile, easily duplicated, alterable/time-sensitive, jurisdiction-spanning, voluminous, format-dependent.
- One line on admissibility requirements (authentic, reliable, complete, legally obtained + §65B/§63 certificate).

D11. Distinguish physical, logical and sparse acquisition.

- Physical: every sector, enables deleted-file recovery.
- Logical: active files only, no deleted data.
- Sparse: selected data + some deleted fragments, middle ground for very large systems.

D12. What are the contents of a forensic report?

- List 6–8 items: case details, authorisation/scope, chain of custody/exhibits with hashes, tools/methods with versions, acquisition + hash verification, findings, analysis/opinion (labelled), declaration/§65B-§63 certificate.

D13. Distinguish between phishing, vishing and smishing.

- Same social-engineering goal; channel differs — email (phishing), voice call (vishing), SMS (smishing).
- One example and one artifact/detection point for each.

D14. Distinguish internal and external attacks.

- Table: access (already have it vs must gain it), detection difficulty (hard vs easier), typical evidence, typical defence (least privilege/DLP vs firewall/IDS/patching).

D15. Distinguish a virus, a worm and a trojan horse.

- Virus: needs host + user execution.
- Worm: self-replicating, no host/user action needed, spreads via network.
- Trojan: disguised as legitimate, does not self-replicate.

D16. What is packet sniffing? How is it carried out on a switched network?

- Definition: promiscuous/monitor mode capturing all traffic.
- On a switch (traffic normally isolated per port): ARP spoofing, MAC flooding, or port mirroring/SPAN to defeat isolation.
- Legitimate use (troubleshooting/IDS) vs malicious use (credential/PII theft from unencrypted traffic).

D17. What is a mobile database and what challenges does mobility create?

- Definition: DB running on/synced with mobile devices under intermittent connectivity/limited power.
- Challenges: replication/sync, conflict resolution, caching/hoarding, disconnected operation, on-device security.
- Note SQLite as the standard on-device engine — forensic payoff.

D18. Distinguish hard and soft handoff.

- Hard (break-before-make) = GSM; soft (make-before-break) = CDMA. One line each on mechanism and network association.

D19. What is frequency reuse? Why are cells drawn as hexagons?

- Definition: reusing the same channel set in non-adjacent cells to multiply capacity.
- Hexagon: tessellates without gaps/overlap, models roughly circular coverage efficiently.

D20. Distinguish IMEI and IMSI.

- IMEI: unique to the handset (hardware); IMSI: unique to the subscriber, stored on the SIM.
- Forensic use of each (device identification vs subscriber/account identification).

D21. What is the order of volatility? Why does it matter in digital forensics?

- List (most→least): registers/cache → network/process state → RAM → disk → remote logs/backups.
- Link to Law of Progressive Change (evidence has a "shelf life"); collect most-volatile data first.

D22. What is drylabbing? State two other common ethical failures in forensic science.

- Definition: reporting results of tests never performed.
- Two more: overstating conclusions ("absolute match"); selective/omitted reporting of exculpatory results.

3.2 Ten-mark questions (13 — Section B style)

D23. Explain the basic principles of forensic science with suitable examples and their relevance to digital evidence.

- List all six (Locard, Individuality, Comparison, Analysis, Progressive Change, Circumstantial Facts).
- One physical example + one digital analogue per principle (see FOR_01 §1.5 table).
- Close with the memory hook and one line tying principles to court admissibility.

D24. Describe the organisation and functioning of forensic science laboratories in India with a diagram.

- Draw the hierarchy: MHA → DFSS/BPR&D/NCRB/NFSU; DFSS → CFSLs + GEQD; State side: State Home Dept → SFSL → RFSL/MFSU.
- One function each for DFSS, CFSL, GEQD, SFSL, RFSL, MFSU.
- Nagaland-relevant note: served by CFSL Guwahati (△ verify current State FSL status).

D25. Explain the different types of spoofing with examples and their defences.

- Table: IP, MAC, ARP, DNS, email, caller-ID/SMS, website/URL spoofing — what is forged, abuse, defence for each.
- Close with the ARP-vs-DNS-vs-IP layer distinction (a common trap).

D26. Describe the common ATM and UPI frauds and how each is investigated.

- List mechanisms: skimming, cloning, trapping, shimming, phishing/vishing/smishing, SIM swap, UPI collect/QR/screen-mirroring scams, money mules.
- For each, name the investigative artifact (CCTV, CDR, UTR/RRN, KYC, device ID).
- Close with the "OTP/PIN never shared" + 1930/cybercrime.gov.in golden-hour point.

D27. Explain SQL injection, XSS and CSRF, and how each appears in web-server evidence.

- One paragraph each: mechanism + forensic footprint (payload patterns in logs; injected script tags; foreign Referer/Origin on authenticated request).
- Close with the XSS-vs-CSRF distinction (script injection vs session riding).

D28. Discuss data acquisition methods (physical/logical/sparse, live/static) and when each is used.

- Two axes: completeness (physical/logical/sparse/targeted) and system state (live/static).
- One line each on trade-offs and typical use-case.
- Close with "modern standard: RAM live-first, then dead-box disk."

D29. Explain the phases of the digital forensic process model with a diagram.

- Draw: Identification → Preservation → Collection → Examination → Analysis → Presentation, with chain of custody running through all.
- One goal + one failure mode per phase.
- Mention NIST's 4-phase condensation and one other named model (Kruse & Heiser / DFRWS).

D30. Discuss forensic report writing and the duties of an expert witness in an Indian court.

- Report qualities (accurate, objective, complete, clear, reproducible, defensible, within competence).
- Report contents (see D12 list).
- Expert duties: primary duty to the court, testify within field, disclose limitations, §45 IEA/§39 BSA advisory status (△ verify), old CrPC §293/BNSS provision for certain government experts (△ verify).

D31. Trace the evolution of cellular networks from 1G to 5G.

- Table by generation: era, switching (circuit→packet→all-IP), key technology, data capability.
- One line each on GPRS/EDGE as 2.5G/2.75G overlays and VoLTE for 4G/5G voice.

D32. Explain mobile database synchronisation and its forensic significance.

- Concept: local storage + replication/sync with server; conflict resolution; caching/hoarding.
- Forensic payoff: SQLite files hold SMS/calls/contacts/chat-app data — name where examiners look.

D33. Discuss ethics in forensic science. What are the common ethical failures and their consequences?

- Code-of-conduct bullets (objectivity, competence, validated methods, documentation, integrity, confidentiality, no overstatement).
- Failures list (drylabbing, overstatement, falsification, selective reporting, testifying beyond expertise) with consequence (wrongful conviction/exoneration, case dismissal, professional sanction).

D34. Enumerate the divisions of a full-service Forensic Science Laboratory and state the functions of each.

- Table: Biology, Serology, DNA, Chemistry, Toxicology, Narcotics, Physics, Ballistics, Documents, Fingerprint, Cyber/Computer Forensics, Photography, Lie Detection, Prohibition/Excise (mention any 8–10 with one function each).

D35. Discuss malware taxonomy and explain why rootkits and fileless malware change the forensic acquisition strategy.

- Table: virus/worm/trojan/ransomware/rootkit/bootkit/keylogger/spyware/botnet/logic bomb/backdoor/fileless/cryptominer/APT — one defining property each.
- Reasoning: rootkits subvert the OS itself → live OS view untrustworthy → dead-box imaging + memory forensics required; fileless malware lives in RAM → RAM capture becomes essential, not optional.

3.3 Fifteen-mark questions (9 — Section C style)

D36. "Physical evidence does not lie." Discuss the principles, significance and limitations of forensic science in the Indian criminal justice system.

- Six principles (brief, one line each) as the foundation.
- Significance: 6–8 points (objective proof, linkage triangle, exculpatory value, reconstruction, identification, deterrence, corroboration, handles no-witness crimes).
- Limitations: backlog, contamination risk, interpretive/expert bias, cost/infrastructure, CSI effect.
- Close with a balanced conclusion — evidence is only as reliable as its collection and interpretation.

D37. Discuss the structure of forensic science services in India from the Ministry of Home Affairs down to the district level, and evaluate the adequacy of this structure for handling cybercrime.

- Full hierarchy diagram (as in D24) plus RFSL/MFSU district layer.
- Adequacy evaluation: strengths (referral system, NFSU training pipeline, mobile units, BNSS mandate) vs weaknesses (backlog, uneven State capacity, shortage of trained cyber examiners, jurisdictional/attribution difficulty in cybercrime).
- Conclusion: structure suits traditional physical evidence better than borderless, fast-moving digital evidence; recommend capacity-building (state cyber labs, NFSU-trained manpower, faster inter-State cooperation).

D38. "Forensic evidence is only as trustworthy as the analyst." Discuss with reference to bias, ethical failures and quality-assurance measures.

- Bias table (confirmation, contextual, expectation, motivational/role, anchoring, reverse-reasoning, observer effect) with one example each.
- Ethical failures list (see D33).
- QA measures: Linear Sequential Unmasking, blind verification, documented SOPs, proficiency testing, accreditation (ISO/IEC 17025/17020 — Δ verify numbers).
- Conclusion tying bias-control to admissibility and wrongful-conviction prevention.

D39. "Attribution is the central problem of cyber-crime investigation." Discuss with reference to spoofing, anonymisation and the classification of cyber crimes.

- Explain why attribution is hard: volatility/remoteness, anonymity (spoofing/VPN/Tor), scale.
- Both classifications of cyber crime (role-based and victim-based) with examples.
- Spoofing family table (IP/MAC/ARP/DNS/email/caller-ID/URL) as the technical mechanism defeating attribution.
- Conclusion: legal process (ISP/platform records), technical correlation (logs, timestamps, hashes) and international cooperation (MLAT) as partial answers to the attribution problem.

D40. Describe the complete lifecycle of digital evidence from identification to presentation, referencing ACPO principles, hashing, chain of custody and the §65B/§63 certificate.

- Six-phase process model diagram.

- Weave in at each phase: ACPO principle invoked, hashing/chain-of-custody action taken, and (at presentation) the §65B IEA/§63 BSA certificate requirement (△ verify exact current section).
- Close with the reproducibility argument (an independent third party should reach the same result).

D41. Discuss the architecture of a cellular mobile network (cells, base stations, MSC/HLR/VLR, handoff) and explain how mobility and switching evolved from 1G to 5G.

- Diagram: hexagonal cells, base stations, frequency reuse, MSC/HLR/VLR roles.
- Handoff types (hard/soft) and which generation/technology uses which.
- Generation table (1G→5G) emphasising the circuit→packet→all-IP switching evolution.

D42. Discuss the various forms of cyber crime targeting individuals, property, government and society, with the applicable legal provisions (verify sections before quoting in the exam).

- Four-bucket table (person/property/government-terrorism/society) with 2–3 examples each.
- Cite the IT Act sections named in FOR_03 for each bucket, flagged △ verify.
- Close with one line on the BNS/BNSS/BSA 2023 overlay replacing IPC/CrPC/Evidence Act.

D43. Explain the digital forensic investigation of a suspected malware/ransomware incident from first response to reporting, integrating order of volatility, imaging, hashing and recovery techniques.

- First-responder actions (secure scene, assess power state, isolate network).
- Order-of-volatility-driven capture: RAM first (fileless/rootkit risk) → live triage → power-down decision (pull-plug vs live-first reasoning) → dead-box disk imaging with write blocker + hashing.
- Recovery: undelete → carving → journal/shadow-copy recovery; note SSD TRIM caveat.
- Report and testimony stage tying back to ACPO and the §65B/§63 certificate.

D44. Compare and evaluate physical, logical and sparse data acquisition together with live and static acquisition strategies, and justify the acquisition strategy you would choose for (a) a powered-off desktop, (b) a running server suspected of an active intrusion, and (c) a locked, powered-on smartphone.

- Definitions/trade-offs table (as in D28).
- (a) Dead-box, physical, write-blocked image — full recovery capability, no volatile risk since machine is already off.
- (b) Live-first: capture RAM/network state/logs before any shutdown decision (server — avoid destructive downtime/data loss); then controlled shutdown or continued live monitoring per ACPO Principle 2, fully documented.
- (c) Mobile-specific: isolate (Faraday/airplane mode), assess AFU/BFU state, use an appropriate mobile acquisition level, document because a hardware write blocker cannot be attached to soldered eMMC/UFS.
- Conclusion: no single "right" method — decision follows from device state, encryption status, and risk of destructive change, always under ACPO Principle 2's competence/documentation requirement.

4. Quick recap — where each syllabus unit is covered in this bank

Syllabus unit	MCQ blocks	Descriptive Qs
TP-I §1 Intro to Forensic Science	A	D1, D3, D23, D36
TP-I §2 Physical Evidence	A	D2, D22
TP-I §3 Scene of Crime	A	(search patterns/chain of custody folded into Block A + D2/D17-style skeletons)
TP-I §4 Computer Hardware	B	(hardware facts recur inside D24/D43 skeletons)
TP-I §5 Windows/Linux/macOS Artifacts	C	D8 (MBR/GPT is TP-II §8 but tool-adjacent)
TP-I §6 Logic/Number Systems	D	— (pure MCQ topic; descriptive coverage lives in the CORE/shared-core notes)

TP-I §7 Computer Networks	E	D25 (spoofing), D16 (sniffing)
TP-I §8 Operating Systems	F	D35 (malware)
TP-I §9 COA	D	—
TP-II §1 Cyber Crime + First Responder	G	D14, D15, D16, D26, D27, D39, D42, D43
TP-II §2 Web Browsers / Email / VM-Cloud	J	(folded into D30/D40 report-writing skeletons)
TP-II §7 Mobile Computing / Mobile Forensics	L	D5 (MFSU relates), D17–D20, D31, D32, D41
TP-II §8 Digital Evidences	H, I, K	D9–D13, D21, D28, D29, D30, D40, D44

Reminder: no official CTSE Computer Forensic past paper exists as of this writing — every question above is a constructed practice item mapped to the official syllabus, not a leaked or recalled paper. Use it to build pattern-recognition (paired traps, table-shaped answers, skeleton-first writing), not to memorise a "final" answer key.